

Ground-Based Aircraft Recorders: Requirements, Challenges, Implementations, and Future Perspectives

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ABSTRACT

The aviation industry's commitment to safety spans across civilian, commercial, and military sectors, with continuous efforts to improve aircraft safety. Despite advancements, fatal accidents still occur, and reliance on the flight data and cockpit voice recorder (FDR/CVR), commonly called the black box, has limitations. This article examines an innovative approach involving the real-time transmission of critical aircraft data to ground-based stations. It introduces the traditional black box, highlights some incidents where it was not beneficial, and then emphasizes the advantages of ground-based recorders over traditional ones, discussing system requirements and challenges. The article evaluates the potential of very high-frequency digital link (VDL) and Iridium satellite systems in handling comprehensive aircraft data in various scenarios. Additionally, it explores employing emerging low earth orbit (LEO) satellite constellations to facilitate FDR/CVR data streaming. The article highlights promising research areas for improving ground-based FDR/CVR systems.

INTRODUCTION

Flight Data Recorders (FDR) and Cockpit Voice Recorders (CVR) play a crucial role in the investigation of air crashes. The FDR is responsible for capturing essential flight data parameters, such as altitude, airspeed, and vertical acceleration, obtained from sensors throughout the aircraft. This data is transmitted to the Flight Data Acquisition Unit (FDAU), which formats it before storing it in the FDR. Currently, the National Transportation Safety Board (NTSB) in the United States mandates monitoring and recording 88 specific parameters by this system [1]. The CVR captures audio from the cockpit, including pilots, air traffic control conversations, and various sounds like warning alarms. These devices have developed significantly, from early models that used photographic film and metal foil to the latest solid-state recorders with increased recording capacity [2].

LIMITATIONS OF EXISTING FLIGHT RECORDERS

Despite the critical role that FDR/CVR recorders play in accident investigations, challenges in their recovery have persisted. These challenges impact the effectiveness of investigations, incur substantial costs, and sometimes endanger the

lives of those involved in recovery operations. Flight recorders' limitations are evident in cases like the most recent January 2024 accident (Alaska Airlines Flight 1282) [3], where their usefulness can be compromised despite the CVR/FDR remaining intact after the accident. In this specific incident, the CVR data was overwritten because the circuit breaker was not pulled after the aircraft landed, illustrating a potential drawback in the design of traditional recording systems. Since 1980, nearly 30 aviation accidents have been documented where the flight recorder was either unrecoverable or damaged, leading to unsuccessful data retrieval attempts [2]. Among these incidents, it becomes evident that traditional FDR/CVR recorders were not consistently reliable. Even in cases where the FDR/CVR recorders were located, the recovery operations were reported to be expensive and time-consuming.

Recovering FDR/CVR from aircraft accidents over water poses extra challenges. The depth of the underwater location necessitates specialized tools like sonar for navigation, and successful diving operations require skilled divers and equipment such as remotely operated vehicles (ROVs). Adverse weather conditions and the instability of the wreckage complicate the secure extraction of FDR/CVR. Another challenge is the limited battery life of the underwater locator beacon (ULB). International incidents may involve complex coordination among various agencies. Commercial aviation accidents from 2009 show instances where FDR/CVR recovery extended to a week or more (Table 1), with some cases taking up to two years to locate the recorders [4].

The exploration of a new real-time flight recording concept is gaining prominence in the aviation industry, driven by substantial advantages over the traditional FDR/CVR system. Cost constraints and time-sensitive recovery procedures associated with FDR/CVR retrieval highlight the need for an alternative. The current recording systems also have limitations in data storage capacity, posing a challenge to obtaining a thorough understanding of accident scenarios. Real-time transmission of aircraft data during emergencies offers a critical advantage, allowing ground-based personnel to provide timely support and potentially prevent accidents or mitigate their impact. Additionally, offsite storage of real-time flight data

Accident Date	Flight Name/Number	Accident Location	FDR Recovery Time (days)	CVR Recovery Time (days)
Jul 2021	Transair Flight 8101	Mamala Bay, Oahu	123	123
Oct 2018	Lion Air Flight 610	Java Sea	3	77
May 2016	EgyptAir Flight 804	Mediterranean Sea	41	41
Dec 2014	Indonesia AirAsia Flight 8501	Java Sea	16	17
Jan 2010	Ethiopian Airlines Flight 409	Mediterranean Sea	14	22
Jun 2009	Yemenia Flight 6266	North of the coast of Grande Comore	60	60
Jun 2009	Air France Flight 447	Atlantic Ocean	679	680

TABLE 1. Some aviation incidents over water from 2009 involving the retrieval of the CVR or FDR taking a week or longer.

could facilitate proactive maintenance, reducing downtime and improving overall aircraft safety.

STATE-OF-THE-ART

Efforts to make FDR/CVR data instantly available by transmitting flight data to ground stations have gained significant attention from researchers and manufacturers over the past two decades, particularly after the Malaysian MH370 flight disaster in 2014. Initial investigations, such as that by [5], explored the feasibility of real-time flight data transmission systems. However, technological limitations hindered implementation at the time. In his thesis [6], the author explored transmitting real-time flight data from aircraft to ground stations using advanced wireless technology, enabling the development of real-time ground-based FDR/CVR recorders. The feasibility of using VDL and the Iridium Satellite System was examined. In [7], the Flight Data Tracker system (FDT) operation is detailed. It operates on the plane server, central, and data servers, sending flight data to a central server either at the origin airport or in the cloud. The central server determines the shortest radio path between origin and destination airports, displaying a list of cloud data servers linked to radio towers along the route. After a handshake with the first data server, the plane server initiates data transmission, forwarding it to the central server upon completion.

Relay communication is a promising candidate as a robust solution for enhancing FDR/CVR data transmission from aircraft to ground stations. Using other aircraft, satellites, or Unmanned aerial vehicles (UAVs) as relays, decode-and-forward (DF) or amplify-and-forward (AF) relaying can extend coverage and improve reliability, particularly in challenging areas such as oceans and remote regions. The work in [8] investigates using aircraft as relays to enhance air-to-ground communication and expand ground station coverage. By leveraging the aircraft's computational power, it aims to reduce transmission costs through optimized DF relaying. Simulation results show improved data rates and reduced costs. In [9], authors propose a software-defined wireless networking framework to utilize inter-aircraft air-to-air radio links, enabling real-time flight data streaming to ground control centers. A time-expanded connectivity graph and a branch and price algorithm were developed to optimize flight data flows transmitted via air-to-air links. The study in [10] examines a UAV-assisted RF/FSO (Radio Frequency / Free Space Optics) communication system using AF relaying with vari-

able gain. This approach, where the UAV acts as a relay, can be applied to data transmission from aircraft to ground stations.

Currently, there are limited options for wireless FDR/CVR systems available in the market. To the best of our knowledge, only two products exist: the Honeywell Connected Recorder (HCR-25) and the FLYHT recorder (AFIRSTM 228). These recorders rely solely on the Inmarsat and Iridium LEO satellite systems, which can be costly. Later, we explore alternative collaborative solutions to address this limitation.

In [2], we presented implementation scenarios and feasibility analyses for real-time ground-based FDR/CVR systems. We also provided an overview of aviation communication and prospects for aviation communication and UAV wireless networks. Beyond the scope of [2], this article further explores traditional recorders' challenges, especially in search processes over water. Additionally, we explore technologies that could enhance ground-based recorders, such as emerging LEO satellite constellations. Going deeper, we discuss ground-based recorders' challenges, introducing new aspects. In addition, we attempt to address future perspectives in research, including integrating Artificial Intelligence (AI) into the evolving space-air-ground network. In the subsequent sections of this article, we outline the system requirements. We then delve into the challenges of implementing real-time flight data streaming. Following that, we propose three potential system implementations. Additionally, we explore communication techniques with the potential to enhance the system. Finally, our conclusions are presented.

GROUND-BASED FDR/CVR RECORDING SYSTEM REQUIREMENTS

Establishing a ground-based FDR/CVR data transmission system that ensures all aircraft FDR/CVR data is accessible at a ground station within a specific region requires addressing several key requirements. These include enabling data transmission from aircraft during flight, ensuring timely and organized data delivery to ground stations, implementing prioritized protocols for emergency transmissions, and optimizing resource usage for efficient data transfer. Research reported in [7] suggests that a data rate of 1800 bps per aircraft is sufficient to ensure real-time transmission of the 88 parameters mandated by the FAA. This article sets this data rate as the minimum requirement for aircraft-to-ground transmission. In situations marked

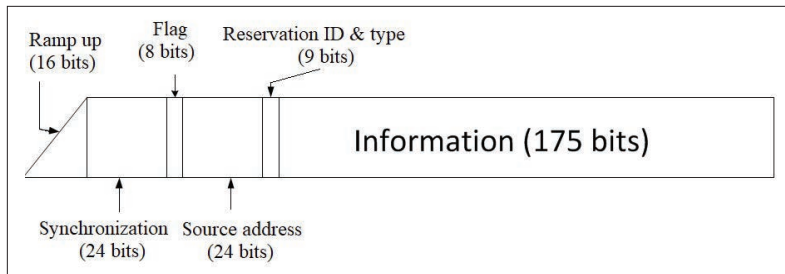


FIGURE 1. First data carrying time slot.

by peak operational periods, particularly in densely populated and urban airspaces like those in the USA, the number of aircraft in the sky can peak at around 5400. Considering this high operational volume, consider a data transmission rate of 1800 bps for each airplane; the system must possess a minimum capacity of 9,720,000 bps. Implementing wireless transmission for aircraft data can be optimized by activating it selectively during significant events instead of continuously transmitting, as this can drain resources unnecessarily. These events, defined by EUROCAE ED-237 criteria, include unusual attitude and speed conditions, collision with terrain, and total loss of thrust or propulsion. This ensures that data transmission occurs only when necessary, preserving resources and enhancing aviation safety protocols. Moreover, the proposed system should offer alternative or cooperative solutions, especially in emergencies or areas with limited connectivity. In such scenarios, the system should be capable of adapting to alternative communication channels, like satellite communication, to ensure data transmission even in challenging environments.

CHALLENGES OF REAL-TIME DATA STREAMING IMPLEMENTATION

As the aviation industry aims to adopt ground-based FDR/CVR capabilities, it encounters significant obstacles. This section explores the intricacies of integrating real-time flight data streaming.

THE COST FACTOR

Despite widespread support for FDR/CVR data streaming, many airlines are hesitant to integrate it due to perceived high costs in an industry marked by narrow profit margins. The aviation sector, facing financial challenges from fuel and government fees, reports a reduced profit of \$2.25 per passenger in 2023, down from \$5.8 pre-COVID [11]. Airlines argue that real-time data transmission would make flights economically unviable. This hesitance is compounded by concerns about the pricing changes from satellite providers for such services, further complicating the decision. This reluctance stems from a misunderstanding of technology, exemplified by systems like FLY-HTStream AFIRS, which minimizes costs by transmitting data only during unusual incidents. We later explore potential implementations to further reduce ground-based FDR/CVR costs. Illustrating cost-effectiveness, had such technology been on MH370, streaming for seven hours would cost around \$3,000 [2], justifiable compared to the extensive expenses incurred in searching for the missing aircraft's recorders.

The aerospace industry faces distinct challenges due to the long lifespans of commercial airplanes, making technology upgrades time-consuming and costly. Retrofitting older aircraft with new technology is particularly expensive and complex. Integrating new technology into precision-crafted aircraft from manufacturers like Boeing and Airbus poses significant challenges. The aviation sector operates as a "long-cycle" industry, characterized by slow rates of technological advancement, in contrast to rapidly evolving sectors like mobile cellular wireless communication systems, where generational shifts occur every decade. Such changes can take three to five decades to materialize in the aviation industry.

PRIVACY CONCERNS

Implementing ground-based FDR/CVR introduces significant privacy concerns, particularly regarding transmitting CVR recordings. CVR may capture personal conversations, necessitating stringent measures to prevent unauthorized access. Despite NTSB's two-decade-long advocacy for cockpit video recording, pilot unions oppose it due to privacy concerns. Streaming data during unusual incidents could offer a potential solution to address privacy concerns and enhance data accessibility.

REGULATORY GAPS

There is no regulatory mandate for live-streaming FDR/CVR data in new aircraft models. Possible regulatory changes are expected to be gradual and complex. Aviation regulatory authorities do not have ongoing global initiatives to establish specific rules for ground-based FDR/CVR products. However, as these products become more available, regulatory considerations may emerge.

Furthermore, regulatory authorities face challenges in assigning new frequencies for such usage methods. Their current focus does not include allocating additional spectrum for live-streaming FDR/CVR data, which could further influence the regulatory landscape for these technologies.

BUFFERING REQUIREMENTS

Data link outages significantly impact the data buffer requirements for ground-based FDR/CVR systems, which must store each aircraft's continuous 1800 bps data stream. Proper design, robust buffering capabilities, high-capacity storage, and efficient compression are essential for maintaining data integrity and ensuring seamless transmission once the link is restored. Chapters 4 and 5 in [2] address buffering requirements for different implementations, providing detailed strategies to optimize storage and manage data during outages effectively. We investigate three real-time ground-based FDR/CVR data streaming approaches in the forthcoming section. In the Iridium Constellation implementation, we examine the impact of introducing buffering on the system's capacity.

IMPLEMENTATIONS

This section introduces three potential ground-based FDR/CVR implementations: VDL Mode 4, the Iridium Constellation, and emerging LEO satellite constellations. The following details elaborate on how these technologies can be utilized for ground-based FDR/CVR implementation.

VDL MODE 4 IMPLEMENTATION

VDL communication systems facilitate data communications by connecting aircraft-based application processes with ground-based counterparts. There are four versions of VDL: VDL mode 0, mode 2, mode 3, and mode 4. These modes are distinguished primarily by their specific multiple access scheme and modulation techniques. VDL Mode 4 is a data communication link employed by aircraft for direct communication with other aircraft or ground stations. This mode employs Self-organized time-division multiple access (STDMA) alongside Gaussian frequency-shift keying (GFSK) on 25 KHz channels, each transmitting at a bit rate of 19,200 bps. The designated frequency band for aircraft communications spans 18.975 MHz and accommodates a total of 760 channels within the range of 118 MHz to 136.975 MHz. In the Medium Access Control (MAC) Layer, the superframe extends for 60 seconds and features a variable number of time slots ranging from 60 to 15,360; in this analysis, the default value is set to 4,500, so the slot time is 13.33 ms long. Transmitting stations can pre-reserve time slots for periodic or unicast transmissions. Additionally, time slots can be randomly accessed when they are not reserved or when access is not predetermined. Each node accessing a single FDMA channel is allotted a maximum of 75 time slots (or 1 second) per superframe for a single transmission. Considering a data rate of 19,200 bits per second, each time slot accommodates a raw data rate of 256 bits (equivalent to 256 bits per 13.33 ms). Regardless of the designated number of time slots per transmission, each transmission is assumed to include 16 bits for ramp-up power stabilization time, 24 bits for ramp-down power decay time, and a 24-bit synchronization field. Consequently, in a scenario where only a one-time slot is employed for transmission, a total or maximum of 192 bits can be transmitted. Figure 1 depicts the structure of the first time slot. To facilitate data transfer at 1800 bps, each aircraft in the system needs to occupy multiple slots within each superframe. Dynamic slot reservation becomes imperative due to the dynamic nature of the process, wherein aircraft frequently alter their operating cells while in transit. Under VDL Mode 4, it is a requirement to reserve every time slot designated for data transmission. The process begins with a transmitting station initiating the reservation by sending three requests in three unused time slots containing information necessary for requesting future data-transmitting time slots. Figure 2 shows the proposed system model for the VDL mode 4 implementations. The requirement for three slots arises due to the utilization of slotted Aloha for each Request To Send (RTS) message, which operates on a random access basis. These three slot requests are expected to result in the allocation of one reservation for time slots each time they are transmitted. Essentially, we operate under the assurance that slots for future transmissions will never be unsecured. For every three RTS messages, an unused time slot must be allocated to receive a Clear To Send (CTS) message before the actual information transmission begins. Furthermore, an extra slot must be reserved for

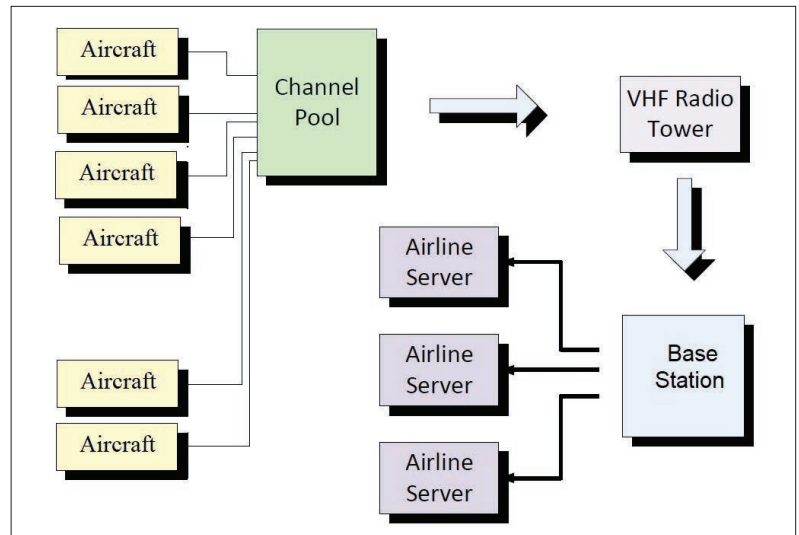


FIGURE 2. Model for the Implementation of VDL Mode 4 System.

the receipt of an acknowledgment (ACK) after the completion of the transmission burst. Each airplane must be capable of accurately transmitting at a speed of 1800 bps. Consequently, CRC (Cyclic Redundancy Check) and FEC (Forward Error Correction) bits are added to the last time slot in every burst. In bursts with more than two slots, the second and intermediate slots can operate at full capacity, accommodating 256 bits of information each. The final time slot, dedicated to carrying 64 CRC bits and 24 bits for power ramp down, is limited to a maximum capacity of 168 bits in each burst. The system estimations necessitate 8 time slots to accommodate 1800 bits of data. The allocation is as follows: the initial slot transmits 175 bits, followed by the subsequent 6 slots, each with a capacity of 256 bits. With an approximate allocation of 64 bits for error checking and correction, the final slot is reserved for 168 bits of user data. The total capacity of these 8 slots is 1879 bits, demonstrating sufficiency to meet the data load requirements for each aircraft.

Capacity Analysis: Each burst in the system consists of 3 RTS slots, 1 CTS slot, 8 data slots, and 1 ACK slot, totaling 13 slots. Each time slot lasts 13.33 ms, and the entire burst duration is 173.29 ms. To ensure complete data transfer, each aircraft must maintain at least one burst per second. In a 1-second superframe, a channel can support bursts for approximately 5 planes, leaving about 133.55 ms or 10 unused time slots. These extra slots can be utilized by base stations to transmit additional data, such as handoff information to aircraft [2]. The spectrum for VDL Mode 4 has 760 channels, accommodating up to 3800 aircraft simultaneously within a single cell. Channels can be reused to prevent interference but are not exclusively dedicated to VDL Mode 4. Other aviation communication systems share the frequency band, limiting the availability of channels for flight data transmissions. Despite this, assuming base station receivers at locations like airports, VDL Mode 4's capacity is sufficient to efficiently transmit data well beyond the maximum number of 5400 aircraft in the USA. More simulation results and additional alternative transmission implementations can be found in Sec 5.3.2 of [2].

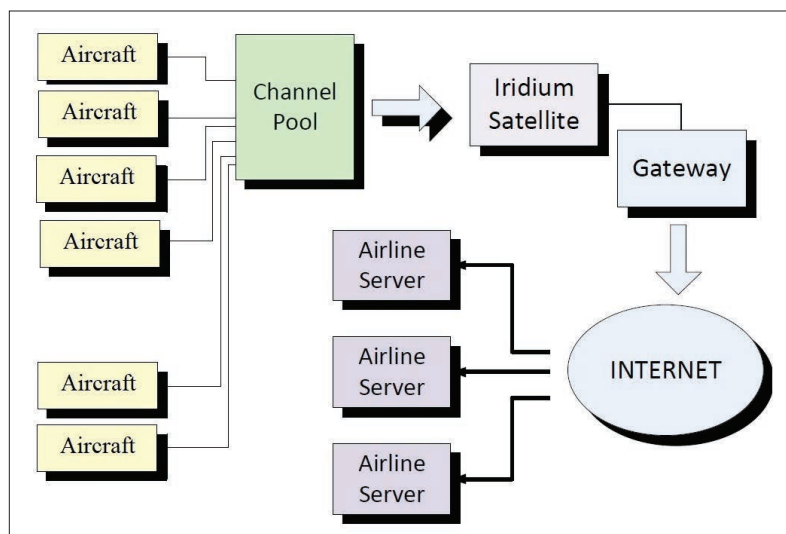


FIGURE 3. System model for Iridium implementation.

VDL System Based Relaying: VDL Mode 4, utilizing STDMA, enables self-organization in aircraft communication by eliminating the need for a master ground station. This is especially advantageous in overcoming limitations related to VHF transmission constraints. The mode allows air-to-air capabilities for data relay between aircraft, addressing regions with limited ground stations. With efficient channel reuse, it offers up to 760 channels in a 400-mile radius, operating autonomously. In scenarios where aircraft are out of range of a VHF ground station, VDL Mode 4 facilitates connectivity through relay aircraft within range. Assuming a maximum of one hop, the relaying aircraft can use two burst frames simultaneously on separate channels, ensuring continuous data transmission. Channels used by relaying aircraft are distinct from those used by out-of-range aircraft, avoiding interference. Transmission slots are reserved through RTS messages for air-to-air communication, with the relaying aircraft confirming via CTS messages. This process allows seamless data transfer between aircraft. Upon entering the range of a ground station, aircraft resume normal burst transmissions after securing reservations.

IRIDIUM CONSTELLATION IMPLEMENTATION

Iridium, a global mobile satellite services provider, operates a robust LEO satellite constellation comprising 66 satellites and spares. The system comprises LEO satellites, Earth stations, system control facilities, and subscriber units.

In this implementation, in Fig. 3, we conduct a case study focusing on the United States to evaluate the data transfer capabilities of the Iridium Satellite system in an area with heavy air traffic. However, the assessment is not confined to this specific region; despite emphasizing a local and regional case study in the United States, the results have global applicability. The system's performance in one area indicates its capabilities in similar regions worldwide, underscoring its consistency and widespread coverage.

Iridium's satellite network is designed to provide extensive coverage, with each satellite having the capability to cover an area approximately 4400 km in diameter. This means a single satellite

can effectively cover the entire United States, with an area of about 9,161,923 km². However, up to four Iridium satellites are strategically positioned to enhance reliability and ensure continuous coverage, especially considering the slight overlap at the edges of their beam coverage areas. Section 4.3 in [2] includes a MATLAB implementation of Iridium satellite coverage.

Capacity Analysis: The Iridium system operates with 240 frequency bands. These frequency bands can be reused every 12 beams, and each satellite contributes 48 beams. This efficient reuse of frequency bands enhances the overall performance and capacity of the Iridium satellite communication system. The collaboration of these four satellites results in a total of 3840 channels. Assuming that FDR/CVR data traffic utilizes 18.75 percent of these channels [6], allowing for the use of 720 FDMA channels. Each FDMA channel can support 8 simplex data transmit slots, totaling 5760 slots for data communication. Each data channel (slot) can transmit at a rate of 2354 bps, resulting in a combined capacity of 13,559,040 bps. This substantial channel capacity ensures that each aircraft can have a dedicated communication channel under average air traffic conditions in the area. At an aircraft data rate of 1800 bps, we can assign each aircraft a fixed slot, resulting in a capacity of 5,760 aircraft.

A more efficient implementation than using a fixed data channel for each aircraft is the fixed slot allocation per aircraft per burst method. In this method, each aircraft is assigned a fixed number of slots during each burst. Each time slot within the 90ms frame can carry 214 bits of user data, so to transfer 1800 bits of FDR/CVR data per second, each aircraft needs an average of 9 slots per second. Since there are approximately 11 frames per second, and each aircraft uses 9 slots per transmission, a single FDMA channel can now support approximately 9 aircraft, an increase from the previous capacity of 8 aircraft per channel. With the system's maximum transmitting capability, this method can support up to 6480 aircraft.

The capacity can be further improved by introducing buffering in the airplanes. Different slot allocations will require different buffering strategies. For example, with 17 slots per transmission, each transmission can transfer 3638 bits. Since each aircraft generates 1800 bits per second, these 17 slots can send two seconds' worth of data. Therefore, each aircraft would buffer data for 2 seconds and use 17 slots in the third second to transmit the data. Another aircraft can access the channel when a buffering aircraft is not transmitting. With approximately 176 slots available in 2 seconds, 10 complete 17-slot transmissions can be sent, allowing 10 aircraft to share the same FDMA channel. Figure 4 illustrates the capacity of a single FDMA channel for the three implementations at different aircraft FDR/CVR transmission data rates.

VHF and Satellite System Cooperation: While both VDL Mode 4 and Iridium individually showcase the ability to manage significant air traffic, challenges in acquiring or affording resources may necessitate their collaborative use. Our proposal involves the cooperation of Iridium and VDL Mode 4 systems for transmitting FDR/CVR data to the ground. Aircraft would dynamically choose between these technologies based on

their specific flight phase. A suggested approach involves using VDL transmissions for flights at lower altitudes and smoothly transitioning to satellite technology for higher altitudes. An alternative approach proposes using Iridium Satellite channels for long-haul flights and VDL for short flights, aiming to be more cost-effective and simpler than altitude-based separation.

EMERGING LEO SATELLITE CONSTELLATION IMPLEMENTATION

Iridium, Globalstar, and other LEO constellations were established before the 2000s. However, the early LEO constellations, consisting of roughly 100 satellites, were limited in scale. Consequently, there was a restriction on the number of satellites visible to each ground station, resulting in a highly constrained design space for satellite-ground topology. Moreover, in earlier attempts, the influence of the satellite-ground topology was confined to a single hop between the satellite and the ground station, resulting in a restricted set of design criteria [12, 13]. The size of LEO constellations has risen rapidly over time, with the number of satellites increasing from a few hundred to thousands. Consequently, the count of visible satellites at ground stations has significantly increased, notably in low and middle latitudes. The expanded scale of the constellation has significantly broadened the design possibilities for satellite-ground topology. This expansion requires ground stations to decide which satellite to connect with. This section delves into the requirements and criteria for implementing an algorithm enabling aircraft to select the optimal line-of-sight satellite for autonomous connection.

The primary goal of designing a real-time communication system for connecting airplanes with satellites and ground stations is to ensure stable and reliable connectivity. Key considerations include establishing links only when airplanes, satellites, and ground stations are within each other's line of sight. This requires a visibility matrix accounting for their respective locations and movements. Maintaining an average topology change interval between all airplane-ground station pairs above a predefined threshold is crucial for network stability. Each aircraft must always maintain at least one link with an LEO satellite, assuming no buffering exists. The optimal path between any given airplane-ground station pair should aim for the minimum hop count while ensuring that the minimum data rate for each airplane-satellite link is at least 1800 bps.

FUTURE PROSPECTIVES

In this section, we describe six important perspectives for the future. The first perspective focuses on A2G (Air-to-Ground) Communication technologies. The future of ground-based FDR/CVR communication in aviation stands on the verge of a revolutionary shift, propelled by promising technologies that can significantly improve data transfer capabilities, diminish latency, broaden coverage, and establish resilient connections between airplanes and ground-based stations. This technological advancement is poised to reshape the efficiency and reliability of FDR/CVR communication systems, opening avenues for advanced data exchange within the aviation industry. Key A2G technologies include mmWave's high-frequency support, mobile edge computing (MEC)

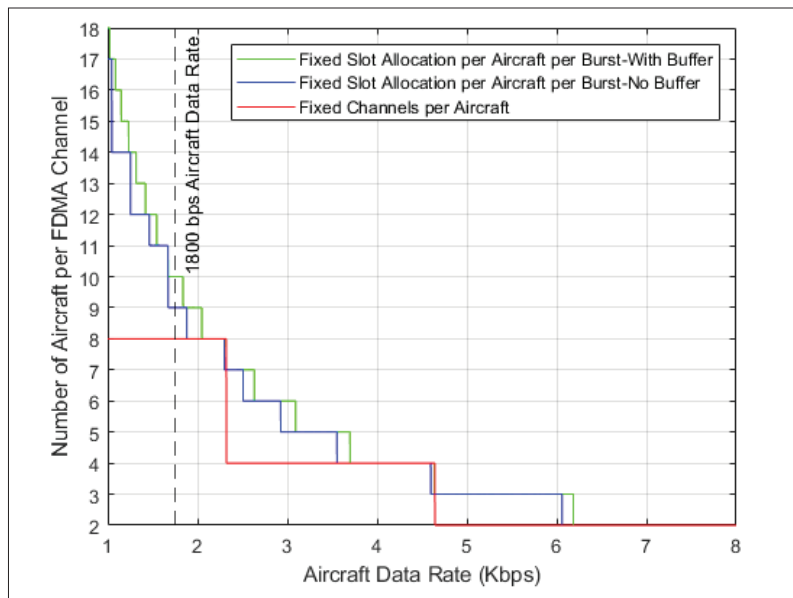


FIGURE 4. Capacity analysis of One Iridium FDMA Channel.

latency reduction, and ultra-wideband (UWB) precise positioning.

Moving to the second perspective, A2A (Air-to-Air) communication technologies are prominent. Several existing A2A, including ACAS (Airborne Collision Avoidance System), ASAS (Airborne Separation Assurance System), L-DACS (L-band Digital Aeronautical Communications System) A2A Mode, and FSO communications, show promise for wireless FDR/CVR data transmission enhancement. ACAS enhances collision risk awareness and resolution advisories for pilots, while ASAS facilitates optimal flight path planning and ensures safe separation distances. Present ACAS and ASAS implementations primarily center around emergency communication between pilots. However, future research could investigate using these systems to enable A2A relaying. L-DACS is a terrestrial system designed to complement VDL mode 2 for advanced services, offering wideband coverage with line-of-sight capabilities. Operating around 1 GHz, it benefits from excellent propagation characteristics. With further investigation and visibility analysis, L-DACS shows promise as a potential candidate for FDR/CVR data transmission. FSO communications offer high-speed connectivity for data transfer, but their reliance on Line of Sight (LOS) alignment makes them sensitive to mobility concerns. Utilizing FSO communication links between aircraft holds promising potential for advancing A2A networks. These technologies collectively represent a spectrum of innovative solutions for enhancing aviation communication systems.

The third perspective is the Aeronautical Ad-hoc Networks (AANET). A significant advancement in FDR/CVR data transmission involves leveraging AANETs. Unlike traditional networks, AANETs enable direct peer-to-peer communication among aircraft in flight, bypassing the need for a central controlling entity. Operating in the 119-137 MHz VHF spectrum, similar to VDL links, AANETs ensure robust Signal to Noise Power Ratio (SNR) with LOS characteristics. In this network, each aircraft functions as a router, routing data packets through air-to-air links and provid-

In this network, each aircraft functions as a router, routing data packets through air-to-air links and providing internet access via satellite or air-to-ground connectivity. The adaptability and versatility of AANETs position them as a promising solution for the complex demands of FDR/CVR data transfer.

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The fourth perspective examines the role of relay communication in enhancing aircraft FDR/CVR data transmission within the broader context of a Space-Air-Ground Integrated Network. In this network, various elements, such as other aircraft, satellites, and UAVs, can be optimized to facilitate reliable and efficient FDR/CVR data transmission. Utilizing these relay nodes can address LOS signal obstructions and improve the overall data transmission architecture. Investigating the feasibility and optimization of these systems represents a promising future research direction for aviation communication networks.

Due to the complexity of aviation environments, the fifth perspective emphasizes the importance of balancing real-time transmission of FDR/CVR data with maintaining high transmission quality. Ensuring real-time data delivery and high transmission quality is critical, especially in dynamic scenarios like adverse weather conditions or congested airspace. Future research should focus on developing strategies and protocols to optimize this trade-off and meet the stringent requirements of aviation communication systems.

Lastly, the sixth perspective emphasizes the Integration of AI. AI could be pivotal in implementing ground-based FDR/CVR systems within a dense Space-Air-Ground future network. AI-driven dynamic routing algorithms will assess data traffic conditions, optimizing data transmission routing dynamically. In scenarios with high data traffic, AI-powered resource allocation becomes crucial as the system intelligently distributes bandwidth and processing resources among ground stations, UAVs, airplanes, and satellites, prioritizing critical data transmission for CVR/FDR.

CONCLUSION

Inspired by the fact that traditional Blackbox has limitations in data recovery, leading to delays and challenges in accident investigations, this article presents real-time ground-based FDR/CVR systems as a solution by transmitting flight data directly to ground stations. This solution improves data accessibility, speeds up investigations, and potentially prevents accidents. While there are challenges like

cost and privacy concerns, the benefits outweigh these issues. The implementations of VDL Mode 4, Iridium, and emerging LEO satellite constellations discussed in this article demonstrate significant potential for deploying this technology. Future research should focus on refining these systems and addressing regulatory hurdles.

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BIOGRAPHIES

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